

CELL-ME: Supervised Cell-Centered Graph Learning for Microenvironment Annotation and Downstream Spatial Analysis

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1 Introduction

Spatial omics technologies provide cell-resolved molecular measurements together with precise spatial coordinates, making it possible to study tissue organization directly *in situ* [1]. A central problem in this setting is microenvironment (ME) annotation: for each cell, the model must infer the local tissue context by combining information from the center cell with the composition and arrangement of its surrounding neighbors, where the center-cell information can be marker expression or cell-type identity depending on the feature setting. Reliable ME labels are important for identifying tumor regions, immune-enriched niches, stromal barriers, tertiary lymphoid structures, and tissue boundaries. However, manual ME annotation remains labor-intensive and difficult to scale across large cohorts or across slices with heterogeneous spatial patterns.

Graph-based modeling is a natural fit for this problem because each cell and its surrounding neighbors can be represented as a local spatial graph. Recent computational approaches have used graph deep learning, graph pooling, and spatial niche discovery to characterize tissue microenvironments or cellular neighborhoods from multiplexed spatial data [2, 3, 4]. Yet high-quality ME annotation requires more than learning generic graph representations. In practical use, the model must transfer across slices, remain robust to changing class balance, and preserve subtle neighborhood distinctions that separate related stromal and immune states. These requirements motivate a task-aligned framework that directly learns ME labels from local spatial graphs while maintaining sufficient representation stability for downstream retrieval, novelty detection, and biological interpretation.

To address this challenge, we introduce *CELL-ME*, an end-to-end supervised graph learning framework for cell-level microenvironment label prediction. *CELL-ME* represents each cell by a 2-hop neighborhood subgraph, augments node features with degree encoding, extracts local structural patterns with a Graph Isomorphism Network (GIN), summarizes each subgraph with attention pooling, and predicts the center-cell ME label using a lightweight multilayer perceptron (MLP). *CELL-ME* is trained with supervised ME-label cross entropy together with a JEPA-SigReg regularization objective: paired graph views are constrained by sign consistency, and graph embeddings are regularized by sliced Gaussian matching following LeJEPA [6]. The resulting model is trained

directly for ME classification, while its learned embeddings remain useful for reference–query retrieval, withheld-class novelty analysis, and downstream biological analysis.

We evaluate CELL-ME on three spatial omics cohorts with curated ME annotations: a colorectal cancer (CRC) CODEX cohort, a triple-negative breast cancer (TNBC) CODEX cohort, and a hepatocellular carcinoma (HCC) spatial proteomics cohort derived from source studies [7, 8, 9]. Across these datasets, we first study standard held-out-slice ME annotation and embedding-based reference–query transfer. We then perform a CRC novelty analysis in which a follicle-associated microenvironment is withheld from the reference split. Finally, we ask a stricter biological question in CRC: although CELL-ME predictions under the limited-label setting are not identical to ground-truth ME labels, can the predicted labels still support the same types of downstream analyses that were originally performed with curated cellular neighborhoods? This final analysis tests whether predicted ME labels are useful not only as benchmark outputs, but also as interpretable units for spatial tissue biology.

Figure 1 summarizes the overall workflow. During training, labeled center-cell subgraphs from seen slices are used to optimize the encoder and classifier with supervised ME-label cross entropy, SigReg view consistency, and LeJEPA sliced Gaussian regularization. During inference, an unseen slice is converted to the same graph representation and passed through the trained model in a single forward pass, producing ME labels for all center cells.

The main contributions of this work are fourfold. First, we formulate ME annotation as direct supervised prediction on cell-centered neighborhood graphs, allowing the model to learn from local spatial context explicitly. Second, we introduce a compact graph architecture that combines degree encoding, GIN message passing, and layer-wise attention pooling. Third, we adapt the LeJEPA JEPA–SigReg principle to graph-level ME embeddings by coupling augmentation-based sign consistency with sliced Gaussian distributional regularization. Fourth, we show that CELL-ME supports not only accurate slice-level annotation and embedding-based retrieval across CRC, TNBC, and HCC, but also CRC-specific downstream analyses that recover known biological patterns despite prediction noise and distinguish biologically cleaner ME labels than a GNN baseline.

2 Method

2.1 Problem Setup and Graph Construction

Let $\mathcal{S} = \{S^{(k)}\}_{k=1}^N$ denote a collection of spatial slices. Each slice $S^{(k)}$ contains n_k cells with spatial coordinates $p_i^{(k)} \in \mathbb{R}^2$, node features $x_i^{(k)} \in \mathbb{R}^D$, and, for training slices, a center-cell ME label $y_i^{(k)} \in \{1, \dots, C\}$. Depending on the feature setting, the node feature can be either a marker panel vector or a one-hot cell-type representation. In the experiments reported here, the downstream benchmarking and CRC analyses are conducted in the cell-type setting, which we denote as **ct** in the result plots.

For each slice we first build a slice-level spatial graph using a 6-nearest-neighbor rule in coordinate

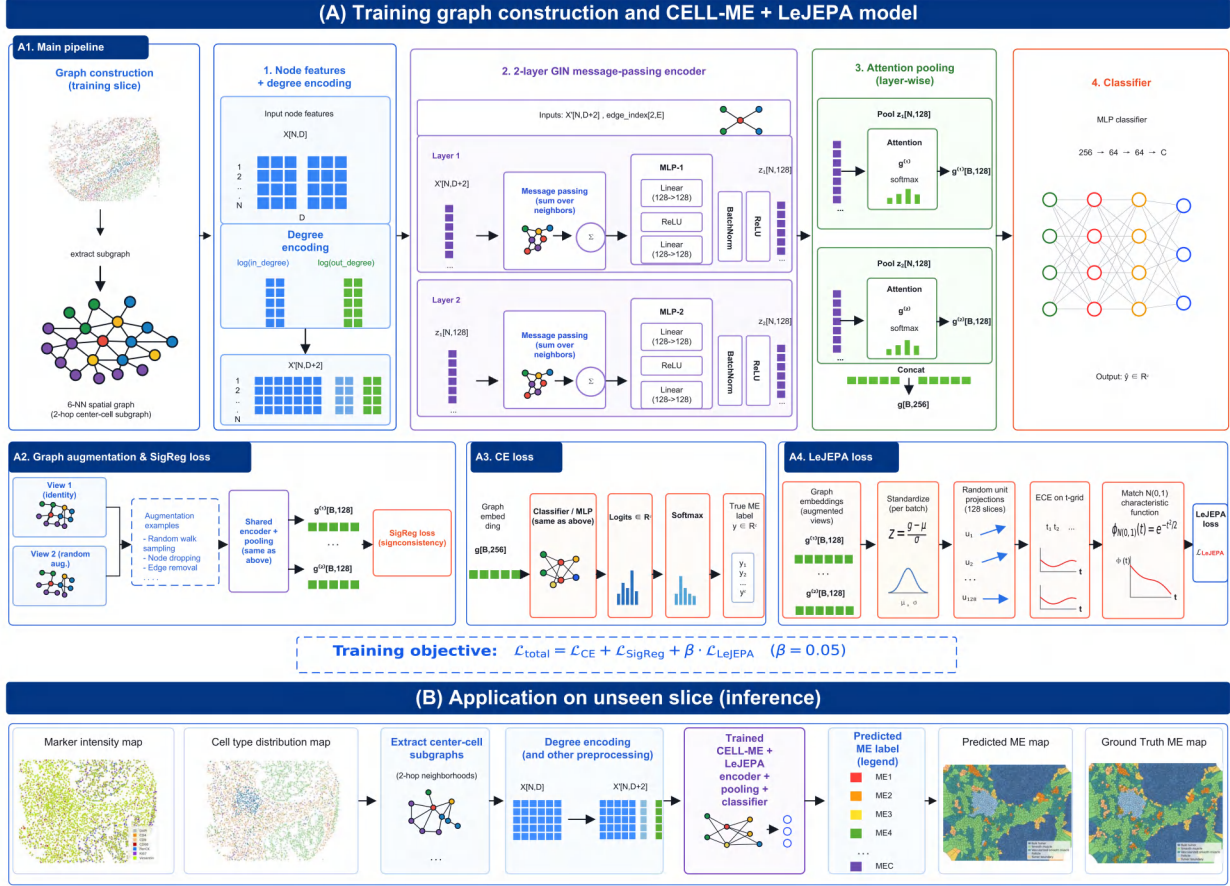


Figure 1: Overview of CELL-ME. Each center cell is represented by a 2-hop cell-neighbor subgraph extracted from a 6-nearest-neighbor spatial graph. The model appends structural degree features, encodes original and augmented subgraphs with a shared GIN encoder, aggregates node information through attention pooling, predicts the center-cell ME label with an MLP classifier, and regularizes the graph embedding with SigReg plus LeJEPA sliced Gaussian loss. The same encoder-classifier pipeline is then applied directly to unseen slices.

space. Formally, the k -th slice is written as

$$G_{\text{slice}}^{(k)} = (V^{(k)}, E^{(k)}),$$

where each node corresponds to a cell and an edge $(u, v) \in E^{(k)}$ indicates that cells u and v are spatial neighbors. Based on this slice graph, we extract a 2-hop induced subgraph for every center cell. For the i -th center cell, the resulting sample is

$$G_i = (V_i, E_i, X_i),$$

where V_i contains the center cell together with all cells reachable within two hops, E_i is the induced edge set, and $X_i \in \mathbb{R}^{|V_i| \times D}$ stacks the node features of all cells in the subgraph. Each graph sample inherits the ME label of its center cell.

This formulation offers two practical advantages. First, it preserves the local tissue context that is most relevant to ME recognition, including both cell composition and spatial arrangement. Second,

it converts a large slice into a collection of aligned graph samples, allowing one shared model to be trained across all cells and slices.

2.2 CELL-ME Architecture

The overall architecture of CELL-ME is shown in Figure 1. Given a cell-neighborhood subgraph G_i , we first enrich each node feature with an explicit degree descriptor:

$$\tilde{x}_j = [x_j \parallel \log(1 + d_j^{\text{in}}) \parallel \log(1 + d_j^{\text{out}})],$$

where d_j^{in} and d_j^{out} denote the in-degree and out-degree of node j in the subgraph after adding self-loops for the encoding step, and $\parallel$ denotes concatenation. This degree encoding gives the encoder lightweight structural information that is not present in the raw panel or cell-type features alone.

The degree-augmented node features are then processed by a two-layer GIN encoder [5]. Let $H_i^{(0)} = \tilde{X}_i$. At the ℓ -th layer, the node embeddings are updated by

$$H_i^{(\ell)} = \text{GIN}^{(\ell)}(H_i^{(\ell-1)}, E_i), \quad \ell \in \{1, 2\}.$$

Each GIN layer uses 128 hidden channels. We obtain one graph summary from each layer and concatenate the two layerwise summaries to obtain a 256-dimensional graph-level representation. Compared with a plain neighborhood averaging scheme, this layered formulation preserves both lower-level and higher-level context within the same subgraph.

Instead of uniform mean pooling, CELL-ME uses attention pooling at each GIN layer. For layer ℓ with node embeddings $h_j^{(\ell)}$, we compute

$$\alpha_j^{(\ell)} = \frac{\exp(a_\ell(h_j^{(\ell)}))}{\sum_{u \in V_i} \exp(a_\ell(h_u^{(\ell)}))}, \quad g_i^{(\ell)} = \sum_{j \in V_i} \alpha_j^{(\ell)} h_j^{(\ell)},$$

and then form the final graph embedding as $g_i = [g_i^{(1)} \parallel g_i^{(2)}] \in \mathbb{R}^{256}$. Here $a_\ell(\cdot)$ is a learnable scoring network for layer ℓ . This layerwise attention pooling lets the model focus more strongly on informative neighbors, such as boundary cells or immune-enriched regions, instead of assigning identical weight to every node.

The graph embedding is passed to a supervised MLP classifier with dimensions $256 \rightarrow 64 \rightarrow 64 \rightarrow C$ to predict the ME label of the center cell:

$$\hat{y}_i = \text{MLP}(g_i).$$

This architecture directly optimizes the end task of center-cell ME-label prediction and avoids the extra representation-transfer step required by the original contrastive formulation. The 256-dimensional graph embedding also serves as the target of the auxiliary SigReg and LeJEPa distributional losses described below. For each cohort, CELL-ME is trained as a closed-set classifier over that cohort’s own ME-label vocabulary, so a model trained on CRC predicts only CRC ME labels, while TNBC and HCC models use their own cohort-specific label spaces.

2.3 Training Objective and Inference

CELL-ME is trained on labeled center-cell subgraphs from seen slices. For each mini-batch, the model uses one supervised classification batch and one paired-view regularization batch. The two graph views are an identity view and a stochastic augmentation sampled from random-walk sampling, node dropping, or edge removal. Both views are encoded by the same GIN-attention network, producing graph embeddings g_i and g'_i .

The supervised term is cross entropy on ME labels:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{i=1}^B \log p(y_i | g_i).$$

The SigReg term is the mean squared sign difference between paired graph-view embeddings:

$$\mathcal{L}_{\text{SigReg}} = \frac{1}{Bd} \sum_{i=1}^B \sum_{r=1}^d (\text{sign}(g_{ir}) - \text{sign}(g'_{ir}))^2.$$

For LeJEPA regularization, graph embeddings in a batch are first standardized by dimension to obtain z_i . We sample S random unit directions u_s and project each embedding to one-dimensional slices $p_{is} = z_i^\top u_s$. For each slice, the empirical characteristic function is

$$\hat{\phi}_s(t) = \frac{1}{B} \sum_{i=1}^B \exp(itp_{is}),$$

and it is matched to the standard Gaussian characteristic function $\phi_0(t) = \exp(-t^2/2)$ on a fixed grid of t values. The loss $\mathcal{L}_{\text{LeJEPA}}$ is averaged over slices, grid points, and the two graph views. In the selected CRC configuration, $S = 128$, the grid has 17 knots, $t_{\max} = 5.0$, and $\beta = 0.05$.

The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{sig}} \mathcal{L}_{\text{SigReg}} + \beta \mathcal{L}_{\text{LeJEPA}},$$

with $\lambda_{\text{sig}} = 1$ and $\beta = 0.05$.

At inference time, each unseen slice is converted to the same 6-nearest-neighbor graph and decomposed into 2-hop center-cell subgraphs. Every subgraph is forwarded once through the trained encoder and classifier, and the predicted class is assigned as the ME label of the corresponding center cell.

3 Model Evaluation and Representation Analysis

3.1 Datasets, Feature Setting, and Evaluation Protocol

We evaluate CELL-ME on three cohorts with curated ME labels: CRC, TNBC, and HCC. Unless otherwise specified, the learned graph model is reported in the cell-type feature setting (ct), where

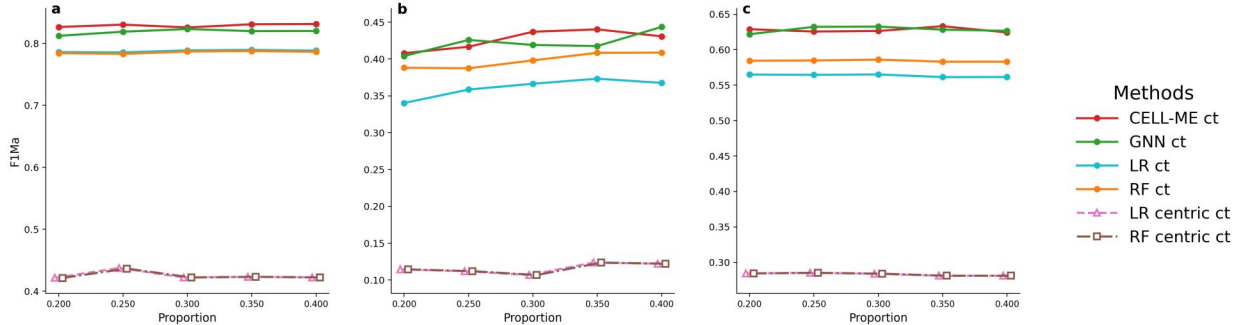


Figure 2: Cross-cohort benchmark on held-out slices. Panels (a)–(c) show macro-F1 performance on CRC, TNBC, and HCC, respectively, under the cell-type feature setting. Each cohort is trained and evaluated within its own label vocabulary under the same slice-wise split protocol, and higher values indicate better ME-label prediction.

each node is represented by a one-hot cell-type vector and the downstream task is to predict the ME label of the center cell. For the standard annotation benchmark, slices are partitioned into seen slices used for optimization and unseen slices used only for evaluation. Only training slices are used for gradient-based optimization, validation slices are used for model selection, and test slices are held out for final evaluation. The graph encoder, attention pooling module, and MLP classifier are then applied directly to each unseen slice without test-time fine-tuning.

3.2 Cross-Cohort Benchmark on Held-Out Slices

Figure 2 reports macro-F1 on held-out slices for CRC, TNBC, and HCC. We use macro-F1 as the main score because ME labels are imbalanced and minority microenvironments should contribute equally to the evaluation. Across the three cohorts, CELL-ME gives strong macro-F1 under the same 6-nearest-neighbor, 2-hop graph construction and cell-type feature setting. In the CRC $\text{prop}=0.20$ setting, the selected model reaches $\text{macro-F1} = 0.8263$ on the test split.

3.3 Embedding-Based Transfer, Retrieval, and Novelty Analysis

To test whether the learned embedding is useful beyond the classifier head, we perform explicit reference–query retrieval at the level of individual cell-centered neighborhood graphs (Figure 3(a)–(c)). The unit of retrieval is not an entire slice or tissue region, but a single graph sample built by taking a center cell, extracting its 2-hop spatial neighborhood, and assigning the sample the ME label of that center cell. Under the $\text{prop}=0.20$ standard split, all train and validation graph samples are pooled into the reference bank, whereas all test graph samples form the query set. This yields 93,939 reference graphs and 159,880 query graphs. After row-wise ℓ_2 normalization, cosine similarity is computed between query and reference representations, and the top- K most similar reference graphs are retrieved for $K \in \{1, 2, 3, 5\}$.

Hit Ratio@ K is defined as a binary success indicator per query graph: if at least one of the top- K retrieved reference graphs has the same ME label as the query graph, the query counts as a hit.

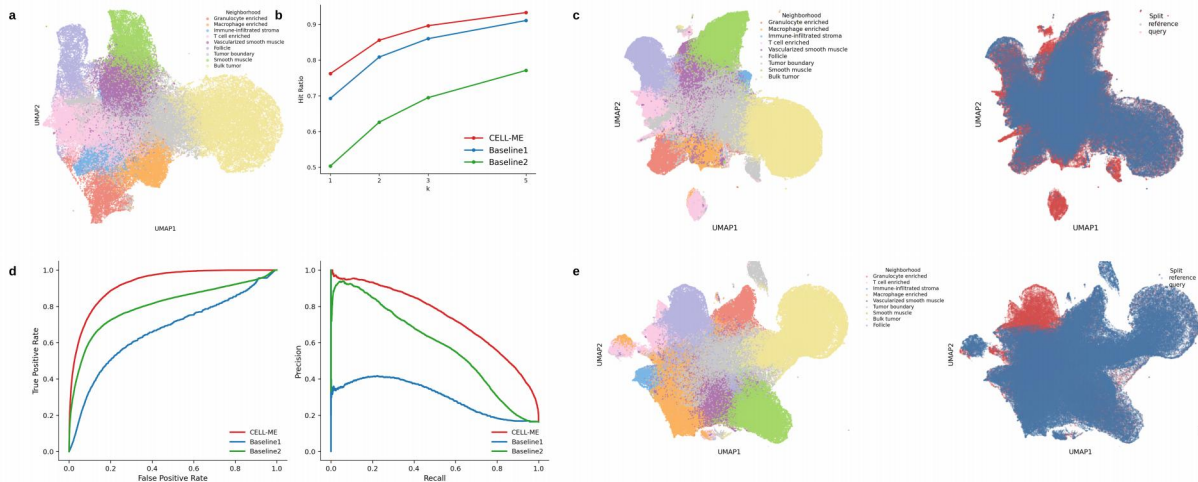


Figure 3: Embedding analysis and novelty detection on CRC. (a) UMAP of the CELL-ME reference embedding under the standard split, colored by center-cell ME label. (b) Reference–query retrieval performance measured by Hit Ratio@ K , comparing CELL-ME with a cell-type composition baseline (Baseline1) and a marker-expression baseline (Baseline2). (c) Joint UMAP of reference and query cells under the standard split, shown by ME label and by reference/query membership. (d) Follicle-held-out novelty detection using ROC and precision–recall curves, where Follicle query cells are positives and non-Follicle query cells are negatives. (e) UMAP of the Follicle-held-out novelty split, colored by ME label and by reference/query membership.

This metric therefore asks a concrete annotation-transfer question: does the retrieved reference neighborhood set contain at least one biologically matching exemplar? For a fair comparison, the same retrieval pipeline is applied to three graph-level representations. CELL-ME uses the learned 256-dimensional graph embedding. The cell-type baseline averages the one-hot cell-type features over all nodes in each graph, which is equivalent to a vector of local cell-type proportions. The expression baseline averages marker/panel features over all nodes in each graph, producing a vector of local mean panel values. All three representations are then evaluated with the same reference/query split, the same row-wise normalization, the same cosine top- K retrieval rule, and the same Hit Ratio@ K metric.

Figure 3(b) shows that the learned CELL-ME representation performs best throughout the retrieval range. Hit Ratio@{1,2,3,5} is 0.7618, 0.8550, 0.8956, and 0.9325 for CELL-ME, compared with 0.6925, 0.8082, 0.8597, and 0.9105 for the cell-type baseline and 0.5030, 0.6259, 0.6946, and 0.7707 for the expression baseline. The advantage is most pronounced at small K , indicating that the learned embedding organizes local microenvironments more sharply than simple neighborhood composition or mean-expression summaries. The UMAP visualizations support the same conclusion: the reference-side embedding already separates most ME classes, and the joint reference–query embedding preserves this organization after held-out slices are added (Figure 3(a),(c)).

We further ask whether the embedding can flag a microenvironment that is absent from the reference ontology (Figure 3(d),(e)). For this analysis, all CRC slices with zero Follicle cells are assigned to the reference side, whereas all slices containing at least one Follicle cell are assigned to the query

side. The zero-Follicle slices are further divided into train, validation, and reference-extra subsets. Only the train subset is used for gradient-based optimization, the validation subset is used for model selection, and the final reference bank is constructed after training from embeddings of all cells in the train, validation, and reference-extra zero-Follicle slices. The query set contains all cells from Follicle-positive slices, so it includes both withheld Follicle cells and known non-Follicle microenvironments from the same tissue sections. This design removes Follicle not only from the reference bank but also from supervised model fitting, making novelty detection a strict held-out-class test rather than an artificial outlier exercise.

After training with zero-Follicle slices and selecting the checkpoint with the validation split, we discard the classifier head for novelty scoring and use the frozen encoder to extract embeddings for all reference-side and query-side cells. Let $R \in \mathbb{R}^{n_r \times d}$ denote the matrix of embeddings from all zero-Follicle reference cells and $Q \in \mathbb{R}^{n_q \times d}$ the matrix of embeddings from all cells in Follicle-positive query slices. Each row is normalized to unit Euclidean norm, yielding $\bar{R}$ and $\bar{Q}$. For the m -th query cell, the implemented novelty score is

$$s_{\text{novel}}(\bar{q}_m) = 1 - \max_{1 \leq j \leq n_r} \bar{q}_m^\top \bar{r}_j,$$

where $\bar{q}_m$ is the normalized embedding of the m -th query cell and $\bar{r}_j$ is the normalized embedding of the j -th reference cell. Because both rows are ℓ_2 normalized, the inner product is exactly cosine similarity, so the score is precisely one minus the best reference cosine match. No top- k averaging, temperature scaling, softmax conversion, or additional z -score normalization is applied.

Positive samples are query cells whose ground-truth ME label is Follicle. Negative samples are all remaining non-Follicle query cells from the same Follicle-positive slices. ROC and precision-recall curves are then obtained by sweeping a threshold directly over the cell-level scores s_{novel} , without any slice-level, region-level, or cluster-level aggregation. Under this setup, CELL-ME achieves a ROC-AUC of 0.9232 and a PR-AUC of 0.7422, compared with 0.6671 and 0.3013 for the cell-type baseline and 0.8049 and 0.5849 for the expression baseline. This setup is more informative than simply checking whether the classifier mislabels withheld cells, because it evaluates whether the embedding space learned from known microenvironments preserves a usable notion of *distance from known biology*. The resulting separation suggests that CELL-ME learns a representation capable of both assigning known labels and flagging coherent unseen microenvironments when the reference ontology is incomplete.

4 CRC Spatial-Ecology Readouts from Predicted ME Labels

4.1 Evaluation Rationale and CRC Reference Framework

The analyses above evaluate CELL-ME as a prediction and representation-learning model. We next ask whether its predicted CRC ME labels are biologically useful as analysis units under the limited-label setting used for downstream analysis. Specifically, Figure 4 uses the prop= 0.20 CRC setting: 20% of labeled data provide the known-label training signal, and downstream spatial-ecology readouts are computed on the remaining 80% prediction set using model-derived ME labels. This question is important because the model is not perfect: under this setting, the predicted labels

cover 253,819 cells from 35 patients and reach a cell-level accuracy of 0.8222 overall, with 0.8374 in CLR patients and 0.8108 in DII patients. Thus, the predicted labels contain nontrivial noise and are not a one-to-one replacement for curated ground truth. The relevant downstream question is therefore whether a noisy predicted label field can still preserve the major spatial-ecological signals that are needed for cohort-level biological interpretation.

To define this downstream evaluation in a biologically grounded way, we adapt the spatial-ecology framework established in the CRC CODEX study by Schürch et al. [7]. That work used cellular neighborhoods (CNs) to connect local cell-type composition, patient-group structure, tissue modules, and T-cell functional states in the colorectal cancer invasive tumor microenvironment. Here, we use CELL-ME predicted MEs as the corresponding spatial units and examine whether they support the same classes of readouts. We also compare CELL-ME labels with GNN-predicted labels on the same prediction subset to ask whether a modest macro-F1 gain translates into more faithful biological semantics. This is not intended as an exact re-annotation of the original CN pipeline; rather, it asks whether supervised graph-predicted MEs retain enough biological structure to act as interpretable spatial units when manual or curated ME labels are only partially available.

4.2 Immune Composition and Patient-Level ME Abundance

We first examine whether the predicted labels recover the known immunological contrast between Crohn’s-like lymphoid reaction (CLR) and diffuse inflammatory infiltration (DII). At the immune-cell composition level, CLR tumors show a pronounced enrichment of B cells, whereas DII tumors contain higher fractions of macrophages and granulocytes (Figure 4(a)). This establishes the expected biological background: CLR is associated with organized lymphoid structure, while DII is associated with a more diffuse myeloid- and granulocyte-rich inflammatory state. Although this panel does not use predicted ME labels directly, it provides the cellular context in which the subsequent predicted-ME analyses should be interpreted.

CELL-ME predicted ME frequencies organize this immune-cell contrast into spatial microenvironment labels (Figure 4(b),(c)).

The strongest patient-group signal is the Follicle ME, whose mean frequency is substantially higher in CLR than in DII (9.70% versus 0.62%, FDR= 0.0138). This direction matches the central CRC observation that CLR tumors contain organized lymphoid follicle-like structures, whereas DII tumors largely lack them. In contrast, DII tumors show non-significant but biologically consistent trends toward higher macrophage-enriched, granulocyte-enriched, and tumor-boundary MEs. The patient-level dot plot adds an important distributional view: Follicle depletion in DII is observed across the cohort rather than being driven by one or two outlier patients, while the myeloid- and granulocyte-associated MEs show broader inter-patient heterogeneity.

4.3 Predicted-Label Semantics and Spatial Coherence

The next question is whether the predicted labels have stable semantic content. The cell-type composition heatmaps show that the major CELL-ME predicted labels are dominated by their expected

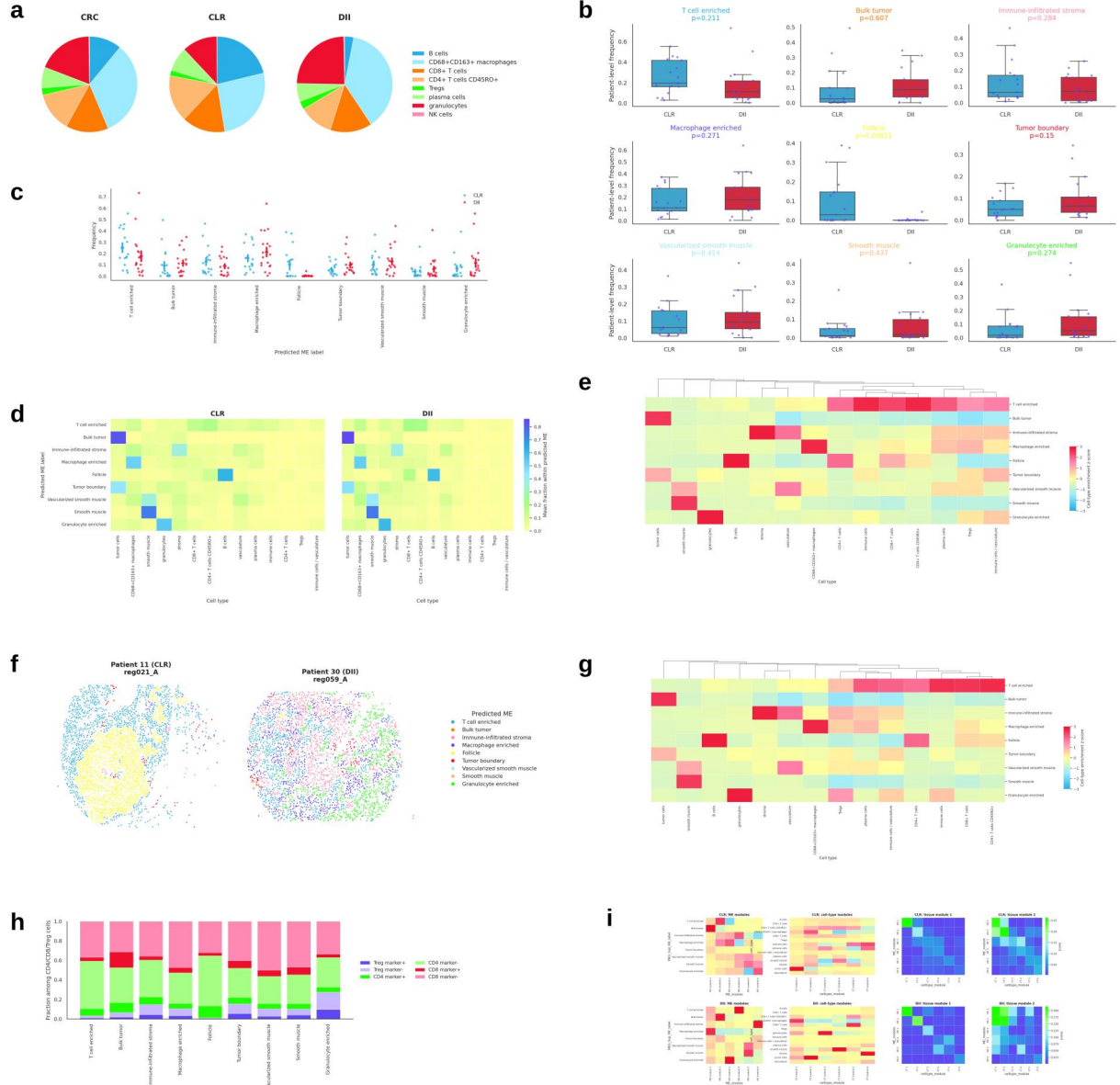


Figure 4: CRC spatial-ecology readouts in the prop= 0.20 known-label setting. Unless otherwise stated, panels use CELL-ME-predicted ME labels on the remaining 0.80 prediction subset. (a) Immune-cell composition in all CRC samples, CLR tumors, and DII tumors. (b) Patient-level frequencies of each predicted ME label in CLR versus DII, with group-comparison p values shown in each panel. (c) Patient-level dot plot of predicted ME frequencies across all ME labels, highlighting between-patient heterogeneity. (d) Cell-type composition within each CELL-ME predicted ME label, shown separately for CLR and DII. (e) Cell-type enrichment z-scores across CELL-ME predicted ME labels. (f) Representative spatial maps of CELL-ME predicted ME labels in a CLR slice and a DII slice. (g) Cell-type enrichment z-scores computed on the same prediction subset using GNN-predicted ME labels. (h) T-cell functional-state composition within CELL-ME predicted ME labels. (i) HOSVD/Tucker-style modules from patient $\times$ CELL-ME predicted-ME $\times$ cell-type tensors in CLR and DII.

lineages in both CLR and DII (Figure 4(d)). Bulk tumor is almost entirely composed of tumor cells, Follicle is B-cell rich, Macrophage enriched is dominated by CD68+CD163+ macrophages, Granulocyte enriched is dominated by granulocytes, and Smooth muscle is dominated by smooth muscle cells. This result is important because downstream analyses would be difficult to interpret if the predicted ME names were disconnected from their cellular composition.

The enrichment heatmap provides a complementary view of the same semantic stability (Figure 4(e)). Whereas the composition heatmap asks which cell types numerically dominate each ME, the enrichment view asks which lineages are selectively overrepresented relative to the overall background. This is where the LeJEPa-regularized CELL-ME labels show a clearer biological advantage over the GNN baseline despite only a modest macro-F1 difference (Figure 4(g)). For Tumor boundary, CELL-ME recovers a mixed boundary profile with tumor-cell enrichment of 1.02 and immune cells/vasculature enrichment of 0.88, whereas the GNN profile is more tumor-skewed and less mixed (0.90 and 0.46, respectively). This better matches the CRC CN-6 description as a tumor plus immune/non-immune interface rather than a pure tumor region [7]. For Granulocyte enriched, both models recover granulocytes strongly (2.79), but the GNN label also pulls in Tregs (1.29) and immune cells (0.74), whereas CELL-ME keeps these signals much lower (0.29 and -0.05). Because the original CN-9 is primarily granulocyte enriched, the CELL-ME label is cleaner and easier to interpret. For Immune-infiltrated stroma, both models capture stromal/vascular structure, but CELL-ME retains a slightly stronger immune-vascular mixture (stroma 2.81, vasculature 1.80, immune cells/vasculature 0.71) than GNN (2.76, 1.83, and 0.46). These comparisons indicate that CELL-ME improves not only the classification score but also the biological fidelity of the predicted ME vocabulary.

Spatial coherence provides a further validation layer. Representative maps show that a CLR slice is dominated by contiguous Follicle and T cell enriched regions, consistent with organized lymphoid architecture (Figure 4(f)). In contrast, a representative DII slice lacks a dominant Follicle structure and instead contains a diffuse mixture of macrophage-enriched, immune-infiltrated stromal, T-cell-enriched, granulocyte-enriched, and tumor-boundary MEs. This distinction is stronger than a simple immune-versus-nonimmune comparison: the predicted maps suggest that CELL-ME captures differences in spatial organization, not only differences in overall immune-cell abundance.

4.4 Higher-Order Tissue Modules and Functional-State Readouts

CELL-ME predicted MEs also retain enough structure to support functional-state analysis (Figure 4(h)). T-cell functional-state composition varies across predicted MEs: granulocyte-enriched and tumor-boundary MEs contain higher fractions of Treg-associated states, whereas Follicle MEs are dominated by CD4/CD8 states with comparatively low Treg marker-positive fractions. This result indicates that the predicted ME labels remain connected to functional immune-state organization, not only to coarse cell-type composition.

We then move from single-label summaries to higher-order tissue organization. Motivated by the tensor-decomposition analysis in the CRC CODEX study [7], we construct patient $\times$ CELL-ME predicted-ME $\times$ cell-type tensors and apply an HOSVD/Tucker-style decomposition separately in CLR and DII (Figure 4(i)). In CLR, the resulting modules separate tumor, immune/myeloid-stromal, and structural compartments. In DII, the dominant axes emphasize macrophage, tumor,

and granulocyte coupling. Because this is an exploratory decomposition in a 35-patient cohort, individual module signs should not be overinterpreted. Nevertheless, the result shows that predicted MEs can support multiway tissue-module analysis, which is a more stringent requirement than recovering marginal ME frequencies alone.

Taken together, Figure 4 evaluates predicted ME labels across a progression of increasingly demanding biological readouts: immune composition, patient-level ME abundance, label semantics, CELL-ME–GNN semantic comparison, spatial coherence, functional-state composition, and multiway tissue modules. The consistent recovery of CLR-associated Follicle/T-cell organization and DII-associated macrophage/granulocyte inflammatory structure indicates that CELL-ME predictions remain biologically meaningful even when cell-level prediction is imperfect. More importantly, the Tumor boundary, Granulocyte enriched, and Immune-infiltrated stroma comparisons show that CELL-ME can produce ME labels that are more faithful to the original CRC CN biology than a GNN baseline, even when the macro-F1 gain is numerically modest. Thus, the predicted ME field is not only a target for macro-F1 benchmarking; it can also serve as a practical substrate for downstream spatial-ecology analysis in settings where complete manual ME annotation is unavailable.

5 Conclusion and Discussion

This study presents CELL-ME as a task-aligned graph-learning framework for microenvironment annotation in spatial omics. By converting each center cell into a local 2-hop neighborhood graph and optimizing the encoder directly for supervised ME prediction, CELL-ME ties representation learning tightly to the final biological objective. CELL-ME combines supervised classification, SigReg view consistency, and LeJEPa sliced Gaussian embedding regularization. The experiments show that this design is useful at three levels: it supports accurate held-out-slice annotation across CRC, TNBC, and HCC; it organizes a reference–query embedding space that remains informative for nearest-neighbor retrieval and novelty detection; and it produces predicted CRC ME labels that remain meaningful for downstream biological analysis despite imperfect cell-level prediction.

The broader implication is that local graph supervision can serve as more than a predictive shortcut. When the embedding is stable under modest graph perturbations and is distributionally regularized against collapse or excessive anisotropy, it becomes a practical substrate for downstream analysis rather than a disposable intermediate feature. That property is especially relevant in spatial biology, where analysts often need to move fluidly between automated labels, neighborhood exemplars, and exploratory inspection of unusual tissue regions. The CRC comparison with GNN-predicted labels further shows why this matters: small differences in macro-F1 can correspond to meaningful differences in whether predicted MEs recover mixed tumor-boundary, clean granulocyte-enriched, and immune-vascular stromal states consistent with the original CRC CN framework.

At the same time, several limitations remain. The current study uses a fixed 6-nearest-neighbor graph and a fixed 2-hop receptive field, which may not capture the most informative spatial scale for every tissue or assay. The main results reported here also focus on the cell-type feature setting, leaving multimodal marker-rich inputs for future comparison. A natural next step is therefore to extend CELL-ME toward adaptive neighborhood construction, richer uncertainty calibration, and cross-platform transfer across assays with partially mismatched feature spaces. Even in its

current form, however, CELL-ME provides a compact and effective framework for supervised ME annotation together with retrieval-, novelty-, and downstream-analysis-oriented use cases.

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